

LAOHEKOU, China

WMO: 572650

Lat: 32.4319N

Lon: 111.7319E

Elev: 423

StdP: 14.47

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.0	28.0	7.8	8.3	40.0	13.1	10.9	39.4	15.5	40.6	12.5	44.7	2.8	70	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	95.6	79.1	93.6	78.3	91.5	77.7	83.2	91.9	81.9	89.9	80.7	88.1	6.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	163.4	89.0	79.8	157.0	87.5	78.7	151.1	86.0	47.8	92.0	46.3	89.9	44.9	88.2	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	13.8	11.4	9.6	DB	21.8	99.9	2.7	1.7	19.9	101.2	18.3	102.2	16.8	103.1	14.9	104.4
(5)				WB	19.9	85.3	2.3	1.2	18.3	86.2	17.0	86.9	15.7	87.5	14.0	88.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.0	38.0	43.0	52.3	63.2	72.1	79.0	82.9	81.2	73.4	64.0	52.2	42.0	(6)
(7)		DBStd	16.49	5.16	6.49	7.17	6.85	5.60	4.88	4.33	5.20	5.71	5.90	6.63	5.02	(7)
(8)		HDD50	942	372	209	57	3	0	0	0	0	0	50	251		(8)
(9)		HDD65	3161	837	617	395	111	9	0	0	6	88	385	713		(9)
(10)		CDD50	5339	1	12	130	398	685	869	1020	968	701	436	116	3	(10)
(11)		CDD65	2084	0	0	3	56	230	419	555	504	257	59	1	0	(11)
(12)		CDH74	20051	0	0	35	446	1820	4075	6226	5298	1797	340	14	0	(12)
(13)		CDH80	8178	0	0	2	111	665	1714	2741	2263	615	67	0	0	(13)
(14)	Wind	WSAvg	4.2	3.7	4.2	4.8	4.7	4.7	4.5	4.3	4.0	3.9	3.8	3.8	3.9	(14)
(15)	Precipitation	PrecAvg	33.3	0.8	1.1	2.0	3.1	3.9	3.2	4.5	4.8	4.5	2.8	1.8	0.7	(15)
(16)		PrecMax	47.4	3.2	4.3	4.5	10.3	9.3	8.9	13.3	11.9	12.6	10.2	5.6	2.4	(16)
(17)		PrecMin	18.7	0.0	0.0	0.2	0.5	1.4	0.4	0.9	0.9	0.4	0.2	0.1	0.0	(17)
(18)		PrecStd	7.7	0.7	0.9	1.1	2.0	1.9	2.0	2.9	2.8	3.2	2.0	1.4	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.3	66.4	78.5	88.4	94.4	97.1	98.6	97.9	94.5	86.4	76.0	63.1	(19)
(20)			MCWB	45.7	51.7	60.2	67.6	69.8	74.7	82.4	80.6	75.5	67.3	59.6	48.2	(20)
(21)		2%	DB	54.2	61.6	72.7	82.9	90.1	94.2	95.7	94.7	89.8	81.3	70.8	58.4	(21)
(22)			MCWB	43.7	49.6	58.4	66.7	70.2	75.1	81.9	80.3	74.0	65.0	57.6	47.0	(22)
(23)		5%	DB	50.5	57.7	68.6	79.1	87.1	91.7	93.5	92.5	86.6	77.8	66.4	54.6	(23)
(24)			MCWB	41.9	47.4	56.7	64.7	69.4	75.2	81.0	79.4	72.0	63.6	55.4	45.0	(24)
(25)		10%	DB	47.1	54.0	64.6	75.3	83.6	88.9	91.1	90.1	83.3	74.2	62.7	51.1	(25)
(26)			MCWB	40.2	45.4	54.2	63.1	68.5	74.5	80.2	78.7	70.8	62.8	54.3	43.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	54.6	65.1	73.3	76.5	81.7	85.2	84.6	80.8	70.5	61.9	51.2	(27)
(28)			MCD B	56.5	62.4	72.7	82.2	85.7	91.3	94.2	94.0	90.1	78.1	71.5	56.9	(28)
(29)		2%	WB	45.0	51.1	61.0	69.5	73.8	80.0	83.8	82.7	77.6	68.2	59.8	48.6	(29)
(30)			MCD B	52.0	59.1	69.4	78.5	83.7	88.8	92.9	90.6	85.2	76.8	67.2	55.2	(30)
(31)		5%	WB	42.9	48.7	58.0	67.0	72.0	78.4	82.6	81.6	75.1	66.4	57.6	46.6	(31)
(32)			MCD B	49.0	55.1	66.1	76.0	81.4	86.4	91.0	89.0	82.2	73.8	63.7	52.8	(32)
(33)		10%	WB	40.9	46.8	55.3	64.5	70.5	76.8	81.3	80.4	72.9	64.7	55.5	44.6	(33)
(34)			MCD B	46.3	52.9	63.5	72.9	79.8	84.4	89.0	87.4	79.8	71.3	61.4	50.0	(34)
(35)	Mean Daily Temperature Range	MDBR	13.5	14.9	16.1	17.3	17.2	15.2	13.0	13.1	13.6	14.4	14.4	14.0		(35)
(36)		5% DB	MCD BR	21.4	22.9	23.7	24.5	24.2	20.4	16.7	17.0	19.5	21.6	22.1	22.3	(36)
(37)			MCWBR	12.7	12.8	11.9	11.2	8.7	7.0	6.3	5.8	6.6	8.1	10.5	12.5	(37)
(38)		5% WB	MCD BR	19.6	19.9	21.0	21.5	18.7	15.9	15.2	14.8	15.8	17.0	18.7	19.4	(38)
(39)			MCWBR	12.4	12.3	11.7	11.6	8.6	7.0	6.6	6.1	7.3	7.8	9.9	11.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.642	0.708	0.701	0.688	0.670	0.724	0.711	0.750	0.728	0.704	0.680	0.624		(40)
(41)		taud	1.537	1.454	1.492	1.555	1.635	1.572	1.630	1.539	1.550	1.531	1.537	1.586		(41)
(42)		Ebn at Noon	172	177	196	208	214	202	204	193	188	176	162	167		(42)
(43)		Edh at Noon	73	88	90	88	82	87	82	88	84	80	73	67		(43)
(44)	All-Sky Solar Radiation	RadAvg	707	813	1171	1397	1475	1520	1469	1336	1083	935	755	672		(44)
(45)		RadStd	108	183	121	141	149	148	201	190	146	148	123	118		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	-0.76	-3.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page